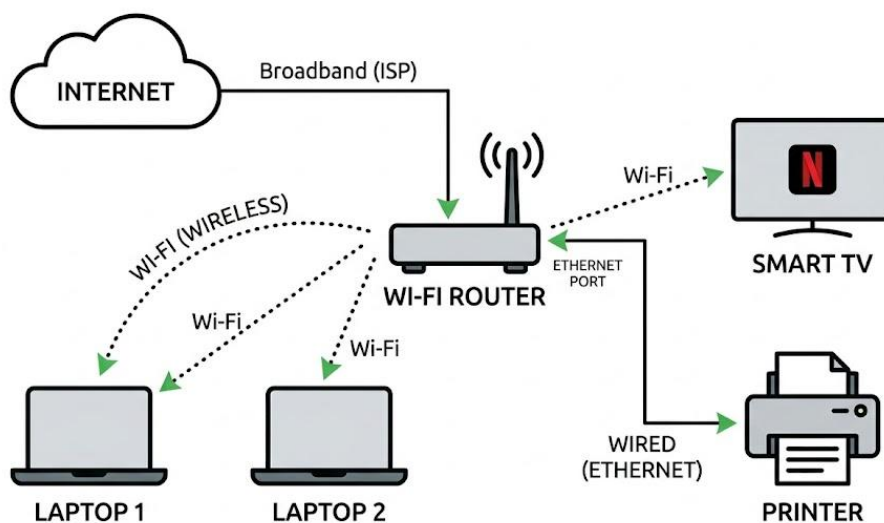


Build a SOHO network

Task 1 :- Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

ANS :-



Task 2 :- List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

ANS :-

Broadband Modem / Router combo : NETGEAR Nighthawk AX1800 (or your ISP-provided router)

Printer : HP LaserJet Pro M15w (Ethernet or Wi-Fi)

Smart TV : Samsung 43" 4K Smart TV

Laptops : Dell XPS 13 & Apple MacBook Air

Ethernet Cable (Wired connection for Printer/TV) : Cat6 Cable (5-10 ft)

Task 3 :- Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

ANS :-

1. Connect & Log In :-

Plug in the router, connect your laptop via Wi-Fi or Ethernet cable, and open a browser. Type 192.168.1.1 (or 192.168.0.1) into the address bar and sign in with the default admin username/password on the label.

2. Change Admin Password :-

Go to System, type a strong new admin password, and save.

3. Set Up SSID (Wi-Fi Name) :-

Go to Wireless Settings and type a unique name in the SSID field (e.g., *MyHomeNetwork*).

4. Enable WPA2/WPA3 Security :-

Set Security Mode to WPA2-Personal (or WPA3), set encryption to AES, create a strong Wi-Fi password, and click Save/Apply.

5. Restart & Reconnect :-

Reboot the router and connect all your devices using your new Wi-Fi name and password.

Task 4 :- Configure IP addresses for all devices in your SOHO network : assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

ANS :-

Device	MAC Address	IP Address	Type
Wi-Fi Router	00:1A:2B:3C:4D:01	192.168.1.1	Gateway
Printer	00:1A:2B:3C:4D:02	192.168.1.10	Static
Smart TV	00:1A:2B:3C:4D:03	192.168.1.20	Static
Laptop 1	00:1A:2B:3C:4D:04	192.168.1.100	Dynamic (DHCP)
Laptop 2	00:1A:2B:3C:4D:05	192.168.1.101	Dynamic (DHCP)

Task 5 :- Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

ANS :-

1. Wi-Fi Turned Off or In Airplane Mode :-

- Fix : Check the laptop settings/keyboard shortcut to turn Wi-Fi back ON and disable Airplane mode.

2. Wrong Wi-Fi Password or Disconnected :-

- Fix : Click the Wi-Fi icon, choose "Forget Network", then reconnect using the correct password.

3. IP Address / DHCP Glitch :-

- Fix : Restart the laptop's network adapter or reboot the Wi-Fi router to get a fresh IP address.